Stored Procedures in SQL

In SQL, stored procedures, functions, and views are powerful database objects that enhance data manipulation, reusability, and security. This document provides an overview of stored procedures and compares them with functions and views.

Stored Procedures

A stored procedure is a precompiled collection of one or more SQL statements or a logical query plan that is stored in the database and can be executed as a single unit. Stored procedures can encapsulate complex business logic and perform various operations, including data modification, data retrieval, and controlling transaction flow.

Key Characteristics of Stored Procedures:

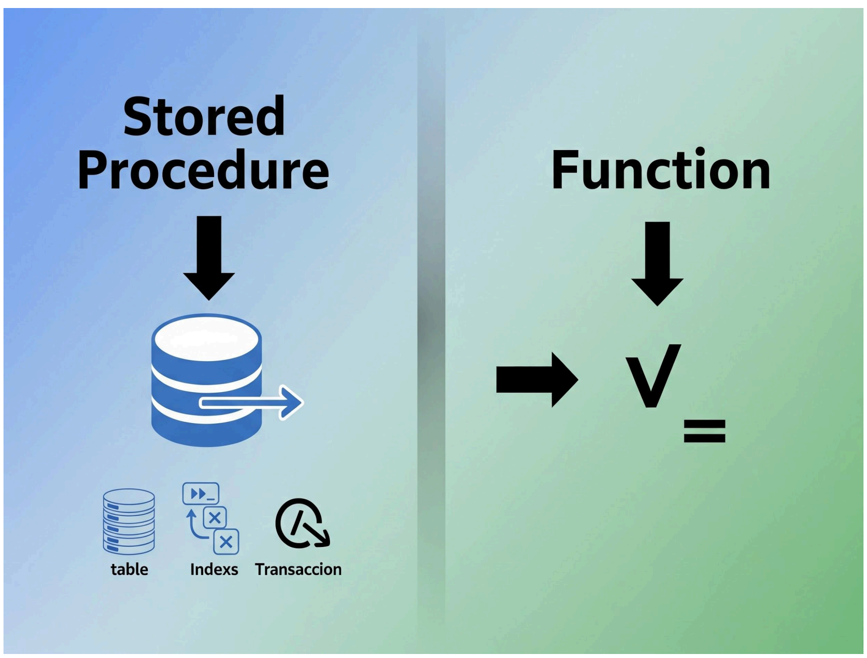
- **Modularity and Reusability:** Stored procedures can be called multiple times from different applications, promoting code reuse and reducing redundancy.
- **Performance:** Once compiled, the execution plan for a stored procedure is stored, leading to faster execution times for subsequent calls.
- **Security:** Users can be granted permission to execute a stored procedure without direct access to the underlying tables, enhancing data security.
- **Reduced Network Traffic:** Instead of sending multiple SQL statements over the network, only the stored procedure call is sent.
- **Parameters:** Stored procedures can accept input parameters and return output parameters, allowing for flexible and dynamic operations.
- **Error Handling:** They can incorporate error handling mechanisms to manage exceptions during execution.

Comparison with Functions

While both stored procedures and functions are precompiled SQL code, they have distinct differences:

Feature	Stored Procedure	Function
Return Value	Can return zero, one, or multiple values (via OUTPUT parameters)	Must return a single value (scalar or table)

Feature	Stored Procedure	Function
Data Modification	Can perform DML (INSERT, UPDATE, DELETE) operations	Cannot perform DML operations (read-only by design)
Call Context	Can be called as a standalone statement	Can be used within SQL statements (e.g., SELECT, WHERE)
Transaction Control	Can manage transactions (COMMIT, ROLLBACK)	Cannot manage transactions
Usage	Executed for specific actions or complex logic	Used to compute and return a value



Comparison with Views

Views are virtual tables that do not store data themselves but represent the result of a stored query. They provide a simplified or customized perspective of the data in one or more base tables.

Feature	Stored Procedure	View
Purpose	Execute a series of SQL statements; perform actions	Represent a virtual table for data retrieval
Data Storage	Does not store data (stores execution plan)	Does not store data (stores the query definition)
Parameters	Can accept parameters	Cannot accept parameters
Modifiability	Can modify underlying data	Can sometimes be updated, but with restrictions (depends on complexity)
Complexity	Can encapsulate complex logic and control flow	Primarily used for simplifying queries and security

When to Use Each

- **Stored Procedures:**
  - When you need to perform complex business logic involving multiple SQL statements and data modifications.
  - For tasks requiring transaction control.
  - To enhance security by abstracting direct table access.
  - To improve performance for frequently executed, complex operations.
- **Functions:**
  - When you need to compute and return a single value (scalar or table).
  - For calculations that can be reused within SQL queries.
  - To encapsulate common expressions or logic that doesn't involve data modification.
- **Views:**
  - To simplify complex queries by predefining a join or selection of columns.
  - To restrict access to certain rows or columns for security purposes.
  - To provide a consistent data interface for applications, even if the underlying table structure changes.

Choosing between a stored procedure, function, or view depends on the specific requirements of your database application. Understanding their individual strengths and limitations is crucial for effective database design and development.